

Sample Question for Unit Test, 2026

Level: BE

Subject : Applied Mathematics

Attempt all questions.

1. Define analytic function. State Cauchy Riemann equations and hence show that it is necessary condition for the function to be analytic.
2. (a) Check the analyticity of $f(z) = |z|^2 = z\bar{z}$.
(b) Check the analyticity of $f(z) = \cos z$. Also, find $f'(z)$ if exists.
3. Is $v = \arg z$ is harmonic? If yes, find a corresponding harmonic function u so that v is harmonic conjugate of u . Also, find the corresponding analytic function $f(z) = u + iv$.
4. Find the bilinear transformation which maps the points $z = 1$, $z = i$, $z = -1$ into the points $w = i$, $w = 0$, $w = -i$. Also, find the image of $|z| < 1$ under this transformation.
5. State and proof Cauchy Integral Theorem. Does Cauchy integral Theorem can be applicable to evaluate the integral $\oint_c \frac{dz}{z^4 + \pi}$, where c is a unit circle $|z| = 1$? If possible, find the value.
6. State and proof the Cauchy Integral Formula. Does Cauchy integral formula can be applicable to evaluate the integral $\oint_c \frac{z^2}{z^4 - 1} dz$, where c is a circle $|z + i| = 1$? If possible, find the value.
7. State the Extension of Cauchy Integral Formula for Derivatives. Evaluate $\oint_c \frac{z^6}{(2z-1)^6} dz$, where c is a circle $|z| = 1$ (counterclockwise).
8. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{z^2-1}{z^2+5z+6}$ when
(i) $|z| < 2$ (ii) $2 < |z| < 3$ (iii) $|z| > 3$ (iv) $1 < |z+1| < 2$.
9. State Cauchy Residue Theorem. Then evaluate the integral $\oint_c \frac{e^z+z}{z^3-z} dz$, where c is a circle $|z| = \frac{\pi}{2}$.
10. Define Z-Transform. State and proof first shifting theorem of Z-transform and using it find the Z-transform of e^{-iat} . Then find $Z(\cos at)$ and $Z(\sin at)$.
11. Solve the difference equation by using Z-transform:

$$y_{n+2} - 4y_{n+1} + 4y_n = 2^n, \quad y_0 = 0, \quad y_1 = 1$$

”ALL THE BEST.”